Practice workbook · The difference of the roots

Using (alpha - beta)^2 = (alpha + beta)^2 - 4alpha*beta to find alpha - beta, and using it in difference-of-squares factorisations such as alpha^4 - beta^4 or alpha^2 - beta^2 to get exact surd values.

12 question parts: 1 worked-example retry and 11 further practice parts.
Name: _____ Class: _____ Date: _____

Practise one question type at a time. Attempt the part inside the red boundary; the rest of the question and any figures are kept for context. Write your working in the answer space.

The first question is a retry of the website worked example, without its solution. Check your answers online with 'Check my answer', or ask your teacher for the answer booklet.

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Type ID: FPM-QUA-08

1. Question 11 · part b.ii · 2 marks

2023 JUN · Paper 02

Worked-example retry · Attempt part b.ii only.

11 The roots of a quadratic equation E are α and β where $\alpha > \beta > 0$

Given that $\alpha - \beta = 2\sqrt{6}$ and $\alpha^2 + \beta^2 = 30$

(a) show that

(i) $\alpha\beta = 3$

(4)

(ii) $\alpha + \beta = 6$

(2)

(b) Without solving E

(i) find the value of $\alpha^4 + \beta^4$

(2)

(ii) find the exact value of $\alpha^4 - \beta^4$

(2)

Given that $\alpha^4 = P + Q\sqrt{6}$ where P and Q are positive integers,

(c) find the value of P and the value of Q

(2)

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1. Question 11 · part b.ii · 2 marks · Working and answer

Working and answer

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2. Question 10 · part b · 2 marks

2019 JUN · Paper 02

Further practice · Attempt part b only.

10 The roots of the equation $x^2 + 3x - 5 = 0$ are α and β.

(a) Without solving the equation, find

(i) the value of $\alpha^2 + \beta^2$

(ii) the value of $\alpha^4 + \beta^4$

(5)

Given that $\alpha > \beta$ and without solving the equation

(b) show that $\alpha - \beta = \sqrt{29}$

(2)

(c) Factorise $\alpha^4 - \beta^4$ completely.

(3)

(d) Hence find the exact value of $\alpha^4 - \beta^4$

(2)

Given that $\beta^4 = p + q\sqrt{29}$ where p and q are positive constants

(e) find the value of p and the value of q.

(3)

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2. Question 10 · part b · 2 marks · Working and answer

Working and answer

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3. Question 10 · part c · 3 marks

2019 JUN · Paper 02

Further practice · Attempt part c only.

10 The roots of the equation $x^2 + 3x - 5 = 0$ are α and β.

(a) Without solving the equation, find

(i) the value of $\alpha^2 + \beta^2$

(ii) the value of $\alpha^4 + \beta^4$

(5)

Given that $\alpha > \beta$ and without solving the equation

(b) show that $\alpha - \beta = \sqrt{29}$

(2)

(c) Factorise $\alpha^4 - \beta^4$ completely.

(3)

(d) Hence find the exact value of $\alpha^4 - \beta^4$

(2)

Given that $\beta^4 = p + q\sqrt{29}$ where p and q are positive constants

(e) find the value of p and the value of q.

(3)

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3. Question 10 · part c · 3 marks · Working and answer

Working and answer

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4. Question 10 · part d · 2 marks

2019 JUN · Paper 02

Further practice · Attempt part d only.

10 The roots of the equation $x^2 + 3x - 5 = 0$ are α and β.

(a) Without solving the equation, find

(i) the value of $\alpha^2 + \beta^2$

(ii) the value of $\alpha^4 + \beta^4$

(5)

Given that $\alpha > \beta$ and without solving the equation

(b) show that $\alpha - \beta = \sqrt{29}$

(2)

(c) Factorise $\alpha^4 - \beta^4$ completely.

(3)

(d) Hence find the exact value of $\alpha^4 - \beta^4$

(2)

Given that $\beta^4 = p + q\sqrt{29}$ where p and q are positive constants

(e) find the value of p and the value of q.

(3)

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4. Question 10 · part d · 2 marks · Working and answer

Working and answer

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5. Question 10 · part e · 3 marks

2019 JUN · Paper 02

Further practice · Attempt part e only.

10 The roots of the equation $x^2 + 3x - 5 = 0$ are α and β.

(a) Without solving the equation, find

(i) the value of $\alpha^2 + \beta^2$

(ii) the value of $\alpha^4 + \beta^4$

(5)

Given that $\alpha > \beta$ and without solving the equation

(b) show that $\alpha - \beta = \sqrt{29}$

(2)

(c) Factorise $\alpha^4 - \beta^4$ completely.

(3)

(d) Hence find the exact value of $\alpha^4 - \beta^4$

(2)

Given that $\beta^4 = p + q\sqrt{29}$ where p and q are positive constants

(e) find the value of p and the value of q.

(3)

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5. Question 10 · part e · 3 marks

2019 JUN · Paper 02

Further practice · Attempt part e only.



Question 10 continued

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Turn over

5. Question 10 · part e · 3 marks

2019 JUN · Paper 02

Further practice · Attempt part e only.

Question 10 continued

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5. Question 10 · part e · 3 marks

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Further practice · Attempt part e only.

Question 10 continued

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(Total for Question 10 is 15 marks)

5. Question 10 · part e · 3 marks · Working and answer

Working and answer

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6. Question 6 · part a · 3 marks

2021 JUN · Paper 01

Further practice · Attempt part a only.

6 (a) Show that $(\alpha - \beta)^2 = (\alpha + \beta)^2 - 4\alpha\beta$

(3)

The quadratic equation $x^2 - 7kx + k^2 = 0$, where k is a positive constant, has roots α and β where $\alpha > \beta$

(b) Show that $\alpha - \beta = 3k\sqrt{5}$

(3)

(c) Hence form a quadratic equation with roots $\alpha + 1$ and $\beta - 1$

Give your equation in the form $x^2 + px + q = 0$ where p and q should be given in terms of k.

(4)

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6. Question 6 · part a · 3 marks · Working and answer

Working and answer

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7. Question 6 · part b · 3 marks

2021 JUN · Paper 01

Further practice · Attempt part b only.

6 (a) Show that $(\alpha - \beta)^2 = (\alpha + \beta)^2 - 4\alpha\beta$

(3)

The quadratic equation $x^2 - 7kx + k^2 = 0$, where k is a positive constant, has roots α and β where $\alpha > \beta$

(b) Show that $\alpha - \beta = 3k\sqrt{5}$

(3)

(c) Hence form a quadratic equation with roots $\alpha + 1$ and $\beta - 1$

Give your equation in the form $x^2 + px + q = 0$ where p and q should be given in terms of k.

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7. Question 6 · part b · 3 marks · Working and answer

Working and answer

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8. Question 7 · part b · 4 marks

2022 JAN · Paper 02

Further practice · Attempt part b only.

7 $f(x) = 2x^2 + px + q$ where p and q are integers
$g(x) = 14x^2 + 37x + 14$

The equation $f(x) = 0$ has roots α and β

The equation $g(x) = 0$ has roots $\frac{\alpha}{\beta}$ and $\frac{\beta}{\alpha}$

Given that $p + q = -4$ where $p > 0$ and without solving the equation $g(x) = 0$

(a) find

(i) the value of p

(ii) the value of q

(9)

Given also that $\alpha > \beta$

(b) show that $\alpha^2 - \beta^2 = -\frac{3\sqrt{65}}{4}$

(4)

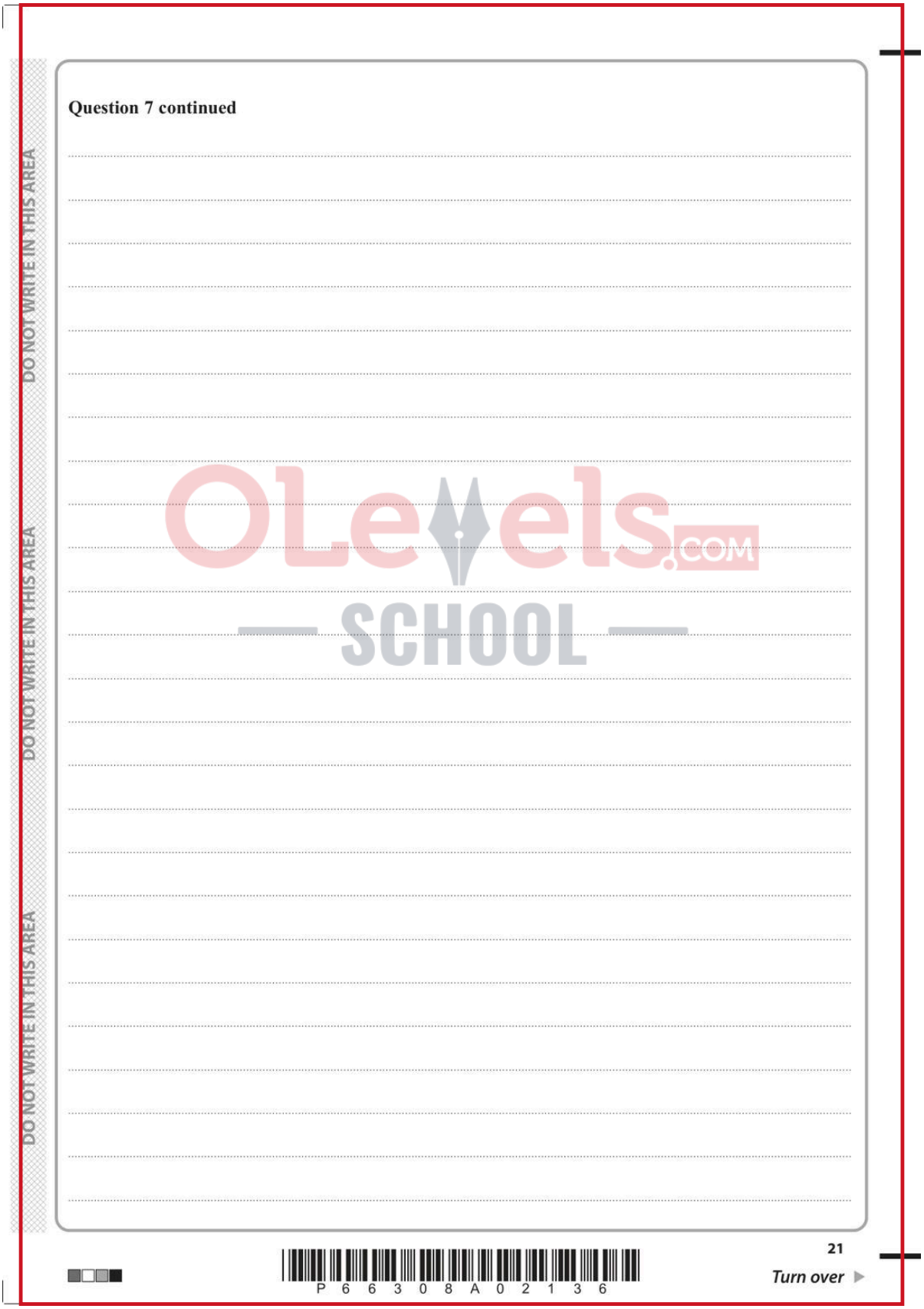
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8. Question 7 · part b · 4 marks

2022 JAN · Paper 02

Further practice · Attempt part b only.



Question 7 continued

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8. Question 7 · part b · 4 marks

2022 JAN · Paper 02

Further practice · Attempt part b only.

Question 7 continued

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8. Question 7 · part b · 4 marks

2022 JAN · Paper 02

Further practice · Attempt part b only.



Question 7 continued

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(Total for Question 7 is 13 marks)

8. Question 7 · part b · 4 marks · Working and answer

Working and answer

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9. Question 5 · part b · 4 marks

2022 JUN · Paper 02

Further practice · Attempt part b only.

5 The roots of the quadratic equation $2x^2 + (6 + 2p)x + 2p = 0$ are α and β

(a) Write down an expression in terms of p for

(i) $\alpha + \beta$ (ii) $\alpha\beta$

(2)

(b) Show that $(\alpha - \beta)^2 = 9 + 2p + p^2$

(4)

Given that $(\alpha - \beta) = 3$

(c) find the possible values of p

(3)

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9. Question 5 · part b · 4 marks · Working and answer

Working and answer

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10. Question 11 · part a.i · 4 marks

2023 JUN · Paper 02

Further practice · Attempt part a.i only.

11 The roots of a quadratic equation E are α and β where $\alpha > \beta > 0$

Given that $\alpha - \beta = 2\sqrt{6}$ and $\alpha^2 + \beta^2 = 30$

(a) show that

(i) $\alpha\beta = 3$

(4)

(ii) $\alpha + \beta = 6$

(2)

(b) Without solving E

(i) find the value of $\alpha^4 + \beta^4$

(2)

(ii) find the exact value of $\alpha^4 - \beta^4$

(2)

Given that $\alpha^4 = P + Q\sqrt{6}$ where P and Q are positive integers,

(c) find the value of P and the value of Q

(2)

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10. Question 11 · part a.i · 4 marks · Working and answer

Working and answer

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11. Question 11 · part a.ii · 2 marks

2023 JUN · Paper 02

Further practice · Attempt part a.ii only.

11 The roots of a quadratic equation E are α and β where $\alpha > \beta > 0$

Given that $\alpha - \beta = 2\sqrt{6}$ and $\alpha^2 + \beta^2 = 30$

(a) show that

(i) $\alpha\beta = 3$

(4)

(ii) $\alpha + \beta = 6$

(2)

(b) Without solving E

(i) find the value of $\alpha^4 + \beta^4$

(2)

(ii) find the exact value of $\alpha^4 - \beta^4$

(2)

Given that $\alpha^4 = P + Q\sqrt{6}$ where P and Q are positive integers,

(c) find the value of P and the value of Q

(2)

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11. Question 11 · part a.ii · 2 marks · Working and answer

Working and answer

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12. Question 11 · part c · 2 marks

2023 JUN · Paper 02

Further practice · Attempt part c only.

11 The roots of a quadratic equation E are α and β where $\alpha > \beta > 0$

Given that $\alpha - \beta = 2\sqrt{6}$ and $\alpha^2 + \beta^2 = 30$

(a) show that

(i) $\alpha\beta = 3$

(4)

(ii) $\alpha + \beta = 6$

(2)

(b) Without solving E

(i) find the value of $\alpha^4 + \beta^4$

(2)

(ii) find the exact value of $\alpha^4 - \beta^4$

(2)

Given that $\alpha^4 = P + Q\sqrt{6}$ where P and Q are positive integers,

(c) find the value of P and the value of Q

(2)

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12. Question 11 · part c · 2 marks

2023 JUN · Paper 02

Further practice · Attempt part c only.

Question 11 continued

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12. Question 11 · part c · 2 marks

2023 JUN · Paper 02

Further practice · Attempt part c only.

Question 11 continued

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(Total for Question 11 is 12 marks)

12. Question 11 · part c · 2 marks · Working and answer

Working and answer

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